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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCTXL100 - TEXTILE SCIENCE

(2024 ADMISSION ONWARDS)

Part A

Multiple Choice Questions

Answer all the questions. Each question carries 1 mark.

1. What happens when cotton fibres are immersed in water?
   Ans: **They absorb water**
2. Identify the distinguishing features of silk fibers under microscopic examination.
   Ans: **Scales**
3. What is a key feature of polyester in terms of durability?
   Ans: **it is highly durable and strong**.
4. Identify natural plant-based fiber?
   Ans: **Cotton**
5. What is the first step in the processing of cotton after harvesting?
   Ans: **Ginning**
6. Identify the smell while wool burns.
   Ans: **Burning of feathers or hair**.
7. List out acrylic fiber commonly used for?
   Ans: **All of the above**
8. Which of the following is a variation of Plain Weave?
   Ans: **Rib weave**
9. Name the fabric that is made using Jacquard weave.
   Ans: **Brocade**
10. Which weave is characterized by a crinkled textured surface?
    Ans: **Crepe weave**

Part B

Fill In the Blanks

Answer all questions

11. The formation of small balls on the surface of fabric is termed as **Pilling or Nubs.**
12. A distinguishing feature of acetate fiber is their **Longitudinal section/ striations**
13. The process of removing seeds from the cotton fibers called **Ginning**
14. Nylon is a **Synthetic** fiber.
15. Nylon 6 is made from a single six carbon substance called **Caprolactam**
16. The rotor speed in rotary spinning affects **Yarn** quality.
17. A higher yarn count indicates a **Thinner/low diameter/fine** yarn
18. Bird's eye weaving comes under the **Dobby** weave category.
19. Lappet weaves use an extra **warp** yarn to create pattern.
20. All laces are characterized by an **Open mesh/ open web** structure and decorative design.

Part C
Answer In One Word or One Sentence.
Answer all questions

21. Which property of cotton fiber allows it to absorb moisture?
    Hydrophilic or Moisture regain
22. What is a common drawback of acrylic fiber?
    Low absorbency
23. Give one major drawback of synthetic fibers?
    Non-biodegradable
24. What is a Acetate rayon? And note down its properties?
    Acetate rayon is a synthetic fabric made from cellulose acetate, a modified cellulose fiber. Lower moisture absorption, higher strength when dry compared to when wet. (write any one property)
25. Which fiber is used for making sports wear?
    Nylon
26. What is the primary purpose of ring spinning?
    Yarn production
27. How is yarn count measured?
    Tex system (Direct system) or cotton count system(Indirect system)
28. What is the primary purpose of using extra warp yarns in weaving?
    For decorative purpose
29. Name another name for Leno weave?
    Gauze/ Duop weave.
30. Classify twill weave based on the angle of the diagonals?
    Right Hand Twill (RHT) and Left-Hand Twill (LHT), regular/medium twill, reclining twill, steep twill (write any two)

Part D
Short Answer Questions
Answer any 5 questions. Each question carries 2 marks.

31. **List down any six properties of Wool.**
    Colour- white, creamy white, to beige, brown and black, **Lustre**- Poor quality wool has higher lustre than good quality wool, Natural **crimp or waviness**, **Lowest strength** compared to other natural fibres, One of the most **Elastic fibers** and very resilient. Therefore, it does not wrinkle, It **drapes** very well, Slowly **absorb** high degree of water, It is an **expensive** fibre.(**write any six**)

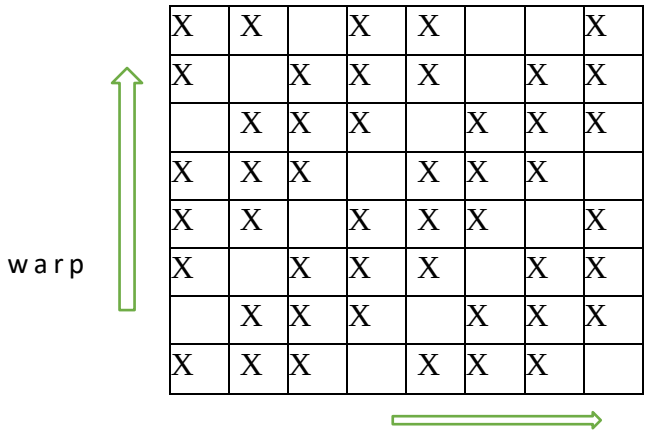
32. **Why are synthetic fibers preferred in Industrial application?**
    Durability: Synthetic fibers are stronger and more durable than natural fibers.
    Cost: Synthetic fibers are cheaper than natural fibers.
    Water resistance: Synthetic fibers absorb less water and dry quickly.
    Elasticity: Synthetic fibers are elastic and don't wrinkle easily.
    Blending: Synthetic fibers blend well with other fibers.

33. **What is the significance of combing in cotton processing?**
    Combing removes shorter, weaker Fibers. It improves the quality of the resulting yarn and fabric. Combed cotton is softer and smoother than non-combed cotton. Combed cotton fabric is more breathable than non-combed cotton. Combed cotton yarn has a higher lustre, or shininess, than non-combed cotton.

34. **Define sericulture?**
Sericulture is the process of cultivating silkworms and extracting silk from them.

35. **Differentiate single yarn and Ply yarn?**
Single Yarn: Single or one-ply, yarns are single strands composed of fibres held together by at least a small amount of twist; or of filaments grouped together either with or without twist.
Ply yarn: Twisting together two or more single yarns of same or different types. Two singles twisted together make a two-ply yarns, whereas three singles make up a three- ply yarn.

36. **Illustrate point paper diagram of a 3x1 Right hand twill?**



X	X		X	X			X
X		X	X	X		X	X
	X	X	X		X	X	X
X	X	X		X	X	X	
X	X		X	X	X		X
X		X	X	X		X	X
	X	X	X		X	X	X
X	X	X		X	X	X	

3/1 RHT TWILL

Part E
Long Answer Questions.
Answer any two questions. Each question carries 5 marks.

37. **Explain the key properties of acetate rayon and discuss its advantages and disadvantages in textile applications?**
Properties:

- Light weight fibre with good drape and silk like aesthetics.
- Low absorbency and therefore resist stains.
- Good resistance to perspiration.
- dries quickly.
- High heat conductivity and provides warmth ie; It conducts heat rapidly making fabrics warm.

Advantage: soft, wrinkle resistant, and fast drying.

Disadvantage: sensitive to heat and can burn easily. It also loses strength when wet, so it should be handled gently and dry cleaned.

38. **Write in detail about Hand spinning?**
Spinning is an ancient textile art in which plants, animal or synthetic fibers are drawn out and twisted together to form yarn.
Oldest Hand spinning machines are:
Distaff and spindle: When spinning with the distaff and spindle, the spinner pulls out from the loose wool or cotton on the distaff, a small piece, and twists the end of it by hand. This finished end is fastened into the notch or slit of the spindle.

Charkha: a small, portable,hand -cranked wheel,is ideal for spinning cotton and other short staple fibers.
Great wheel: The fibre is held in the left hand and the wheel slowly turned with the right.

39. **Discuss the following-swivel weave, lappet weave, spot weave?**
Swivel Weave: Swivel weave is produced by the use of **extra filling yarns.** The yarns used in each pattern are wound on quills and placed in shuttles, which are placed at each point where the design is to appear. This particular weaving process permits weaving in different colours in the same row, because each design has its own shuttle. The design created is the same on both the front and the back side. The fabric has small geometrical patterns and used for apparel and furnishings.

Lappet Weave: Lappet weave introduces one continuous additional yarn in a way that it creates designs over predetermined portions of the basic fabric. One continuous additional yarn is carried on the back of the fabric from one design to the next. The floating thread is cut and the ends are secured. These fabrics are superior to fabrics made with a swivel weave.
Spot Weave: Spot weave is manufactured by the use of either extra warp or filling yarns. The yarns are inserted in the entire length or width of the fabric in predetermined areas and the pattern is formed using either dobby or jacquard attachments. When not in use, the extra yarns float across the back of the fabric. This makes the Uncut spot pattern or Unclipped. At times, when small and widely spaced patterns are made, the floating yarns are out from the back side of the fabric and this is known as the Cut spot yarns or clipped.

40. **Explain physical, chemical and microscopic test of silk?**
Physical properties of silk:

a) Fine smooth and even fabric. It is white in color with very high luster.
b) Medium weight fiber.
c) Thinnest natural fiber and very strong.
d) Retain its shape and resist wrinkles due to its elastic properties.
e) Excellent drapability.
f) Absorbs moisture readily.
g) It is a non-conductor of heat and therefore desirable for winter wear

Microscopic test:

- L.S: Under the microscope the silk fiber looks like a smooth glass rod.
- C.S: Smooth triangular cross section

Chemical properties:

- Alkaline solutions cause the silk filament to swell.
- Silk is degraded more readily by acids; Concentrated sulfuric and hydrochloric acids, especially when hot, dissolves silk.

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